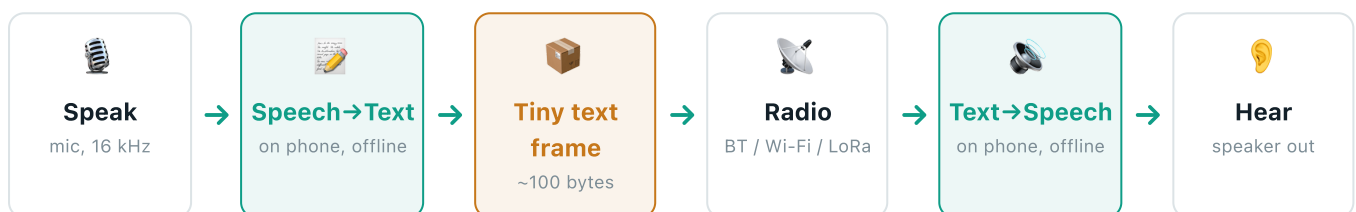


Talk on one phone → hear it on another, with no internet and no towers

Voice is too heavy to send over a satellite, LoRa or HF radio link. So iTantra sends light **text** on the wire and rebuilds the **voice** at both ends — everything runs on the phone itself, fully offline, in 10 Indian languages.

THE CORE IDEA

Instead of squeezing heavy audio through a tiny radio pipe, the app turns your speech into a short line of text (about **100 bytes** — one sentence), sends that text over the radio, and the receiving phone reads it back out loud in a natural voice. **Speech-to-Text** on the way in, **Text-to-Speech** on the way out.



WHAT HAPPENS ON EACH PHONE

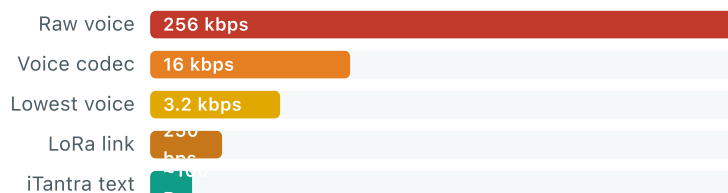
SENDER Turning voice into a tiny message

- Microphone captures your voice
- A small "voice detector" ignores silence to save battery
- Speech-to-Text writes down what you said, word by word
- The sentence is compressed and stamped with a checksum, ready to send

RECEIVER Turning the message back into voice

- The frame arrives and is checked for errors
- Duplicates are dropped; a receipt (ACK) is sent back
- Normal messages become a voice note; ALERTs ring an alarm
- Text-to-Speech reads it out loud in the chosen language

WHY SEND TEXT INSTEAD OF VOICE?



A long-range radio link carries only about **250 bits per second**. Real voice needs far more than that — it simply **cannot fit**. But a sentence of text fits easily, **with room to spare**. Speech-to-Text and Text-to-Speech make that text-only channel feel just like a voice call.

THREE WAYS TO CARRY THE MESSAGE

A Phone to phone

10–100 m · the demo

Phone ↔ Phone

Bluetooth / Wi-Fi Direct

B Radio bridge

2–5 km · long haul

Phone ↔ LoRa))) (((LoRa ↔ Phone

866 MHz · ≈250 bps

C Shared gateway

borrow one radio

Phones → Gateway → Radio

ALERTs get priority

Work distribution across the team

Three members drive the AI models (80% of the score), **Jai** builds the LoRa radio hardware, and **Naitik** builds the Android app. The frozen text-frame lets all three groups work in parallel.

THE THREE GROUPS

3 members AI & MODELS Speech-to-Text · Text-to-Speech · VAD	Pick, shrink and lock the best offline models for each language — this is 80% of the score. - Benchmark STT/TTS per language (accuracy, speed, size) - Quantize models to int8 so they run on cheap phones - Silence detection (VAD) to save battery - Integrate models into the app & measure live latency
Jai LORA MODULE ESP32 + SX1276 radio	Build the long-range radio bridge that carries the text frames 2–5 km. - ESP32 + LoRa firmware (Arduino / RadioLib) - Phone ↔ board link over Bluetooth / USB serial 866 MHz India band, ≈250 bps, error handling - Gateway mode + ALERT priority; live demo
Naitik KOTLIN APP Android · Jetpack Compose	Build the Android app: audio, screens, and moving frames between phones. - UI (Compose), mic/speaker, background service - Build the text frame + checksum, send & receive - Bluetooth & Wi-Fi Direct transport (Backends A & B) - Metrics screen; wire the AI models into the app

HOW THE 3 AI MEMBERS SPLIT THE WORK

AI · MEMBER 1 Speech-to-Text lead Benchmarks & quantizes the STT models, measures word-error-rate per language. owns Accuracy (40%)	AI · MEMBER 2 Text-to-Speech & VAD Picks the voices, tunes speed & size, keeps the app light and fast. owns Latency + Efficiency	AI · MEMBER 3 On-device integration Plugs the models into the app and manages memory. Pairs closely with Naitik. owns the model↔app bridge
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5-WEEK PLAN AT A GLANCE

W1	Benchmark models · prove the toolchain
W2	Quantize + lock models
W3	End-to-end voice, phone ↔ phone
W4	LoRa bridge + gateway + alerts
W5	Polish · results · demo

THE SHARED CONTRACT — THE TEXT FRAME

ver	type	src	dst	seq	payload	crc
4b	4b	8b	8b	8b	text	16b

Same frame over Bluetooth, Wi-Fi and LoRa. **type** = NORMAL / ALERT / ACK · **dst** = 0xFF means broadcast.

Freeze this in Week 1. Once fixed, the AI team, Naitik and Jai can build separately and only meet at integration.